POI-based Traffic Generation via Supervised Contrastive Learning on Reconstructed Graph

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Abstract. Traffic flow generation problem under realistic scenarios has raised more and more attention in recent years. This problem aims at generating traffic flow without using historical traffic data. Since road network and Point of Interest (POI) data can provide a more comprehensive picture of traffic patterns, most previous methods use both or either of them to generate traffic flow. However, roadnet graph in real-world has bias and abnormal structure, which will influence the performance of traffic generation. Previous traffic generation models directly receive real-world roadnet graph with map-match POI data as input and then use an end-to-end loss for training, which could not model the complex relationship between POI and traffic in a proper way. Different from prior methods, we propose a novel POI-based **T**raffic **G**eneration model via **S**upervised **C**ontrastive learning on **R**econstructed graph, termed as **TG-SCR**, which combines POI data and road network data to generate the distribution of traffic flows. Our model has two novel modules: a graph reconstruction module and a POI supervised contrastive module. The graph reconstruction module includes a k-NN graph builder and a k-NN graph aggregator, which is used to reconstruct the original roadnet graph into a k-NN graph and reform POI feature. The contrastive module is used to model the relationship between POI feature and traffic flow. Extensive experiments show that our model outperforms other baseline methods on four real-world datasets.

Keywords: Traffic Generation · Contrastive Learning · Graph Reconstruction.

1 Introduction

Recently, traffic generation problem has become an increasingly popular research topic due to the growing importance of intelligent transportation systems and the availability of large amounts of traffic data. In recent years, deep learning models have also been applied to traffic prediction tasks with promising results. These models can capture spatial dependencies between traffic variables and learn complex patterns in high-dimensional traffic data. In this paper, our primary focus is

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directed toward the "Traffic Generation Task", a distinct undertaking divergent from the "Traffic Prediction Task". The fundamental distinction between these two tasks is that traffic prediction models predict future traffic patterns based on historical traffic data. In contrast, traffic generation models generate the future traffic patterns by leveraging non-traffic data, such as Points of Interest (POI) data or check-in data. To illustrate, consider the task of predicting traffic flow for the next week. While traffic prediction models rely on historical traffic data or equivalent formatted data, traffic generation models only need static data to simulate an authentic environment.

Despite the rapid development of deep learning models in traffic generation domain, most of previous traffic generation models face two challenges. The first challenge is real-world road network graphs often deviate from the elemental principle that "neighbor nodes exhibit similar features". For example, sidewalks in real-world road networks are usually built to connect two areas with completely different POI distribution around, such as commercial and residential areas. The second challenge is the perceived inadequacy of end-to-end loss employed in previous traffic generation models. A predominant approach in prior models involves the imposition of constraints on final predictions through an end-to-end paradigm, typically employing Cross-Entropy loss or Mean Squared Error loss, but do not explicitly consider the hidden representations. This will lead to a severe overfitting problem and reduced accuracy when dealing with more complex non-traffic data.

In this paper, we propose a POI-based traffic generation model which combines POI data and road network data to generate the distribution of traffic flows, thus successfully solving the problem of generating traffic flows in areas without historical traffic data. Our main contributions can be summarized as below:

- **Graph Reconstruction Module.** To solve the first challenge that some pairs of nodes connected by edges in real-world roadnet graphs are not similar in the distribution of POI data around, we add a graph reconstruction module including a ***k*-NN Graph Builder** and a ***k*-NN Graph Aggregator** to reform the original roadnet graph into a k-NN graph and revise the original POI feature. By doing so, our model eliminates biases and anomalies in the real-world roadnet graph, therefore mitigating the negative impact of the message passing mechanism.
- **POI Contrastive Module.** To solve the second challenge that end-to-end loss is hard to constrain the mapping between POI distribution and traffic flow distribution with significant difference, we add a supervised contrastive loss in our model to learn a regression-aware representation by contrasting POI feature embedding against other nodes in a batch based on their target distance. By doing so, our model explicitly holds the former similarity relationships between samples to optimize the representation for the traffic generation task.

– **Evaluation on Real-World Datasets.** Under extensive experiments on four real-world datasets, we show that our model provides consistent boosts in traffic generation performance.

2 Related Works

2.1 Deep Learning Models without Graph

With the development of deep neural networks, many researchers try to apply different deep learning methods to solve the traffic generation problem. Researchers first use RNN-based models such as LSTM[18] to capture the time dependence of traffic flow. In order to consider spatial correlations in traffic flow, researchers also started to use CNN-based models such as UrbanFlow[17]. Afterwards, hybrid models such as SRCN[14] are widely used to solve the traffic prediction task. However, these deep learning models have poor generalization performance and are difficult to use to predict traffic flow under real-world datasets.

2.2 Graph Neural Network Models

In recent years, there has been an increasing enthusiasm in the domain of deep learning for Graph Neural Networks (GNNs). GNNs have emerged as a new and powerful approach for handling graph-structured data, which makes it useful in solving traffic prediction task. In order to capture the dynamic relationship between traffic flow and other influence factors, models such as AGCRN[1] and DGCRN[8] are proposed. However, under real-world datasets, the roadnet graph which GNN-based models research on may have bias, resulting in inaccuracy of traffic flow prediction.

3 Preliminary

Suppose a city-level road network graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$ is given, where n_i denotes the i-th node of the graph and e_j denotes the j-th edge. $\mathbf{G}$ carries a node-level POI features namely $\mathbf{X}$. For each node in $\mathbf{G}$, x_i represents a 24-dimensional Point of Interest (POI) feature vector, where each dimension corresponds to the number of one category of POI points within a 1km radius around n_i. Similarly, y_i represents a 24-dimensional traffic vector, where each dimension corresponds to the number of arrival points within a 1km radius around n_i for a one-hour time slice.

The traffic generation task can be defined as below:

Theorem 1 (Traffic Generation Definition). *Given a roadnet graph* $\mathbf{G}$ *along with its POI feature* $\boldsymbol{X}$ *that has been map-match on* $\mathbf{G}$*, the objective of traffic generation task is to generate a one-day traffic flow* $\boldsymbol{Y}$*. Suppose* $f(\cdot)$ *is a model that maps POI feature to traffic, this traffic generation task can be formulated as:*

$$\boldsymbol{Y} = f(\boldsymbol{G}, \boldsymbol{X}) \tag{1}$$

4 Methodology

4.1 Overview

To generate traffic flow, we propose a novel POI-based traffic generation model via supervised contrastive learning on reconstructed graph, called **TG-SCR**. TG-SCR has two modules: a graph reconstruction module and a POI contrastive module, focusing on structural view and POI contrastive view respectively. The first graph reconstruction module has two components: a k-NN Graph Builder and a k-NN Graph Aggregator. k-NN graph builder is used for converting the original roadnet graph into a k-NN graph, while the k-NN Graph Aggregator is used for aggregating POI feature to generate reforming feature. In the POI supervised contrastive module, we propose a supervised contrastive loss with traffic feature as soft label and then use it to train the residual encoder. The overview of TG-SCR is shown in Figure 1.

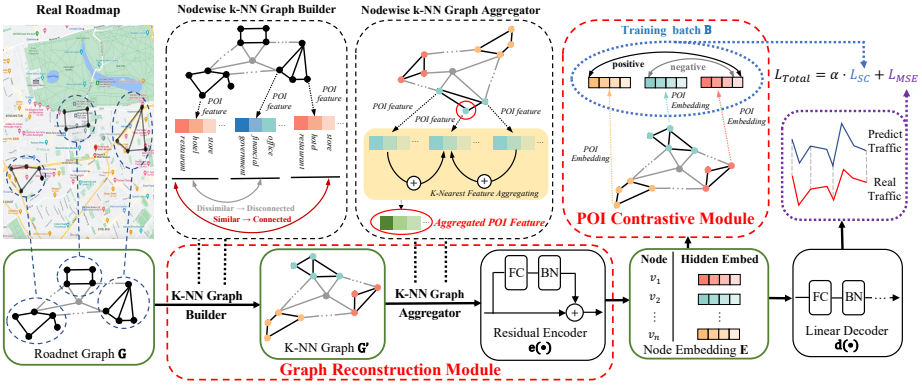


Fig. 1. The overview of TG-SCR. TG-SCR includes a graph reconstruction module and a POI contrastive module. The graph reconstruction module includes a k-NN graph builder and a k-NN graph aggregator. The POI contrastive module models the relationship between POI feature and traffic flow.

4.2 Graph Reconstruction Module

***k*-NN Graph Builder** The k-NN graph builder converts the original roadnet graph G into a k-nearest neighbor graph G'. Assume the i-th node is denoted as n_i with its POI feature X_i and traffic flow Y_i. Each dimension of X_i or Y_i is denoted as x_i^d or y_i^d ($d \leq 24$). For the k-NN graph building step, we first compute the L_2 distance l between the anchor node and other nodes (suppose the j-th node):

$$l_{i,j} = ||x_i - x_j|| = \sqrt{\sum_d \left(x_i^d - x_j^d\right)^2} \tag{2}$$

Suppose the L_2 distances between n_i and other nodes constitute a set $\mathbb{L}_i$. We select k nodes with minimum L_2 distance and connect them pairwisely in the newly formed k-NN graph. These k nodes will be reformed into the neighbor nodes of n_i in k-NN graph. The set of these nodes is denoted as $\mathbb{K}_i$. This step can be formulated as:

$$\mathbb{K}_i = \{n_j \in N : |\mathbb{L}_i \cap (-\infty, l_{i,j})| < k\} \tag{3}$$

***k*-NN Graph Aggregator** After that, a k-NN Graph Aggregator is applied to transform the raw POI feature X into aggregated POI feature X' by adding the nearest POI feature in G'. The aggregating operation can be formulated as below:

$$X'_i = X_i + \sum_{j \in \mathbb{K}_i} X_j \tag{4}$$

4.3 POI Contrastive Module

Suppose in a training batch B with N nodes, the randomly selected anchor node in B is denoted as n_i, its traffic and POI embedding are denoted as y_i and v_i respectively. The predicted traffic of this node is denoted as $\hat{y}_i$. We first select c nearest nodes in B based on their traffic distance and treat them as positive set $\mathbb{P}$, then the rest nodes in B are classified to negative set $\mathbb{N}$. The procedure of generating positive set and negative set can be formulated as below:

$$\mathbb{P} = \{n_j \in B : |Y \cap (-\infty, ||y_i - y_j||)| < c\} \tag{5}$$

$$\mathbb{N} = \{n_k\}, k \neq i, k \in B, k \notin \mathbb{P} \tag{6}$$

Based on the positive set defined in Eq (5) and the negative set defined in Eq (6), our POI supervised contrastive loss L_{SC} can be formulated as:

$$\mathcal{L}_{\mathrm{SC}} = -\frac{1}{N} \log \frac{\sum\limits_{j \in \mathbb{P}} \exp\left(\mathrm{sim}\left(\boldsymbol{v}_i, \boldsymbol{v}_j\right)/\tau\right)}{\sum\limits_{k \in \mathbb{N}} \exp\left(\mathrm{sim}\left(\boldsymbol{v}_i, \boldsymbol{v}_k\right)/\tau\right)} \tag{7}$$

where τ represents the temperature coefficient. In L_{SC} loss, any two nodes can be thought of as a positive pair or a negative pair, depending on their similarities in traffic flow. We define positive and negative samples in a relative way, based on the distance between their traffic and that of anchor node.

4.4 Loss Summary

In addition to the contrastive loss mentioned above, we define another loss to describe the error between predicted and real traffic flow from a global perspective. This loss is used to measure the mean square error between predicted traffic $\hat{Y}$ and real traffic Y. It can be formulated as below:

$$L_{MSE} = MSELoss(\hat{Y}, Y) = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2 \tag{8}$$

The final loss combines the L_{SC} term and L_{MSE} term:

$$L_{Total} = \alpha \cdot L_{SC} + L_{MSE} \tag{9}$$

where α is a hyper-parameter.

To alleviate the gradient vanishing problem during training procedure, we simply train the L_{MSE} for the first five hundred epochs, and then add L_{SC} into training. The training procedure of TG-SCR can be expressed as below:

Algorithm 1 Training procedure of TG-SCR

Input: Roadnet graph G with map-matched POI feature X, encoder $e(\cdot)$ and decoder $d(\cdot)$ parameterized with θ

Build k-NN graph G' from G based on Eq (3)
Get Aggregated POI feature X' from X based on Eq (4)
Compute POI embeddings: $E' = e(X')$
repeat
    Sample a training batch B including N nodes
    Define positive set $\mathbb{P}$ and negative set $\mathbb{N}$ based on Eq (5) and Eq (6)
    Compute L_{Total} based on Eq (9)
    Back Propagation
    Update θ based on Eq (7)
until $L_{Total} < \epsilon$

Output: Predicted traffic $\hat{Y}$

5 Experiment

5.1 Experiment Setup

Datasets We evaluate our model on four real-world datasets of Futian, Nanshan, Shanghai and Beijing. In these four datasets, nodes and edges in the graph are roadnet junctions and roads extracted from OpenStreetMap. Node features consist of POI data, which are queried from Baidu Map with API tools. The traffic data of Futian, Nanshan, Beijing and Shanghai is downloaded from [12], [15, 16] and [9]. For POI data, we divide it into twenty-four categories. For traffic data, we split the one-day trajectory data using our-hour as grain size.

Baseline Methods We compare our model with three categories of methods. The first category includes several traditional machine learning methods used to solve regression problems, such as RandomForest (RF)[2], XGBoost[3] and SVR[4]. The second category is methods that based on Graph Neural Networks, such as GCN[7], GraphSAGE[5], GAT[11] and GIN[13]. The third category includes other models that are used for generating traffic flow based on POI data, such as DeepFlowGen (DFG)[10] and DeepCrowd[6]. The results of the comparison between our model and baselines are shown in Table 1. Code is available at `https://github.com/PoXZ0803/TG-SCR`.

Table 1. Performance comparison between TG-SCR and other baseline methods on four real-world datasets. Results show that our model outperforms other baseline methods in all four real-world datasets. All experiments are repeated five times, and the mean and standard deviation (±) are reported.

Method	Futian			Nanshan		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
RF	64.28	52.83	35.13%	74.72	30.66	24.88%
XGBoost	76.13	62.01	36.37%	67.13	36.81	26.76%
SVR	262.91	157.56	64.06%	100.18	53.59	51.84%
GCN	173.57 ± 4.17	102.46 ± 2.61	$49.76\% \pm 3.54\%$	92.57 ± 0.93	66.50 ± 0.43	$25.38\% \pm 1.39\%$
GraphSAGE	102.64 ± 3.04	74.62 ± 2.56	$42.94\% \pm 2.59\%$	90.18 ± 1.00	63.59 ± 0.46	$24.73\% \pm 0.91\%$
GAT	92.47 ± 1.57	68.47 ± 0.47	$39.83\% \pm 0.82\%$	71.48 ± 1.31	31.09 ± 0.55	$21.41\% \pm 1.57\%$
GIN	79.33 ± 1.11	58.92 ± 0.54	$40.97\% \pm 0.60\%$	65.25 ± 0.62	29.59 ± 0.28	$21.08\% \pm 0.79\%$
DFG	82.39 ± 0.46	63.64 ± 0.37	$38.61\% \pm 0.65\%$	72.46 ± 0.57	36.47 ± 0.30	$23.35\% \pm 0.59\%$
DeepCrowd	92.46 ± 1.38	67.54 ± 0.70	$38.45\% \pm 0.86\%$	74.36 ± 0.37	36.46 ± 0.23	$23.72\% \pm 0.34\%$
TG-SCR	$\mathbf{53.19 \pm 0.40}$	$\mathbf{41.03 \pm 0.21}$	$\mathbf{30.06\% \pm 0.69\%}$	$\mathbf{51.56 \pm 0.42}$	$\mathbf{27.43 \pm 0.24}$	$\mathbf{19.05\% \pm 0.29\%}$

Method	Beijing			Shanghai		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
RF	9.03	4.56	17.62%	6.62	3.02	13.04%
XGBoost	14.52	7.33	29.24%	7.81	4.50	18.89%
SVR	19.12	10.16	36.34%	25.05	13.57	46.84%
GCN	20.46 ± 1.46	13.49 ± 0.84	$56.20\% \pm 1.95\%$	18.34 ± 0.87	11.52 ± 0.33	$42.71\% \pm 3.73\%$
GraphSAGE	22.56 ± 1.36	14.22 ± 0.83	$53.37\% \pm 1.86\%$	14.36 ± 1.31	9.64 ± 0.83	$41.69\% \pm 5.31\%$
GAT	18.92 ± 1.30	12.38 ± 0.56	$33.27\% \pm 2.02\%$	9.44 ± 1.03	6.82 ± 0.95	$28.59\% \pm 3.51\%$
GIN	10.29 ± 0.99	6.22 ± 0.40	$24.68\% \pm 2.42\%$	6.57 ± 0.17	3.79 ± 0.06	$21.06\% \pm 1.42\%$
DFG	3.48 ± 0.16	2.29 ± 0.06	$9.55\% \pm 0.21\%$	2.39 ± 0.04	1.74 ± 0.02	$9.54\% \pm 0.60\%$
DeepCrowd	10.29 ± 1.04	6.22 ± 0.46	$24.86\% \pm 2.89\%$	6.57 ± 0.09	4.25 ± 0.03	$16.63\% \pm 0.92\%$
TG-SCR	$\mathbf{1.47 \pm 0.07}$	$\mathbf{1.10 \pm 0.05}$	$\mathbf{5.12\% \pm 0.14\%}$	$\mathbf{1.67 \pm 0.03}$	$\mathbf{1.26 \pm 0.02}$	$\mathbf{7.31\% \pm 0.83\%}$

5.2 Experiment Results

The results of the comparison between our model and baselines are shown in Table 1. On all of these four real-world datasets with different scales, TG-SCR outperforms other baseline methods in all three evaluation metrics: RMSE, MAE and MAPE. The experiment results show that for all the real-world datasets with different data scale, our model achieves the best performance. Notably, Table 1 shows that some GNN-based methods like GCN perform badly in generating traffic. This is mainly due to the fact that these methods are influenced by abnormal structures in the real-world roadnet graph. GNN's message passing mechanism leads to errors where the predicted traffic of some specific area (e.g. hospital, school) are severely influenced by some more distant areas. TG-SCR overcomes this problem by reconstructing the original roadnet graph and thus avoiding the flaws brought by message passing.

5.3 Ablation Study

As discussed in Section 4.1, TG-SCR includes two novel modules: a graph reconstruction module (denoted as GRM) and a POI contrastive module (denoted as SCM). We perform ablation study to examine the effect of these two modules in TG-SCR. To evaluate, we consider two variants of TG-SCR: **(1)** TG-SCR

without graph reconstruction module (abbreviated as TR-SCR w/o GRM). **(2)** TG-SCR without graph reconstruction module and POI supervised contrastive module (abbreviated as TG-SCR w/o GRM & SCM). Experiment results on Futian and Nanshan dataset are shown in Table 2, from which we could observe that TG-SCR has performance gain over two variants, proving the significance of the proposed graph reconstruction module and POI contrastive module.

Table 2. Performance comparison between TG-SCR and its variants. All experiments are repeated five times, and the mean and standard deviation (±) are reported.

Method	Futian			Nanshan		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
TG-SCR	**53.19 ± 0.40**	**41.03 ± 0.21**	**30.06% ± 0.69%**	**51.56 ± 0.42**	27.43 ± 0.24	**19.05% ± 0.29%**
TG-SCR w/o GRM	55.67 ± 0.42	42.63 ± 0.21	31.28% ± 1.66%	52.72 ± 0.61	**27.19 ± 0.23**	19.28% ± 1.08%
TG-SCR w/o GRM & SCM	92.46 ± 1.38	67.54 ± 0.70	38.40% ± 0.86%	72.46 ± 0.57	36.47 ± 0.42	23.58% ± 8.36%

5.4 Parameter Sensitive Analysis

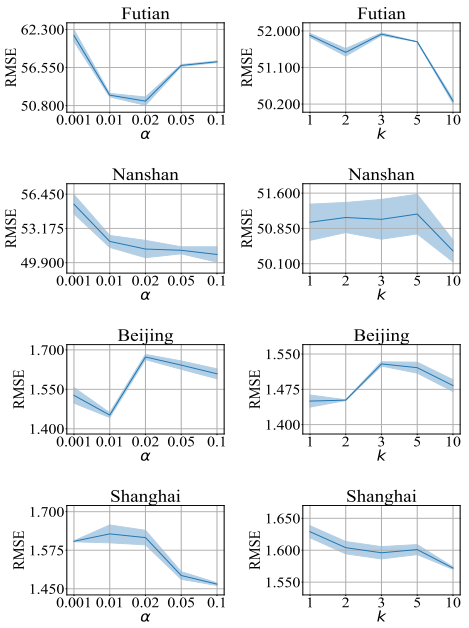


Fig. 2. The results of two parameter sensitivity analysis on four real-world datasets. The corresponding RMSE results are respectively shown using line charts. The blue shaded area in each line chart indicates the error range of the corresponding standard deviation.

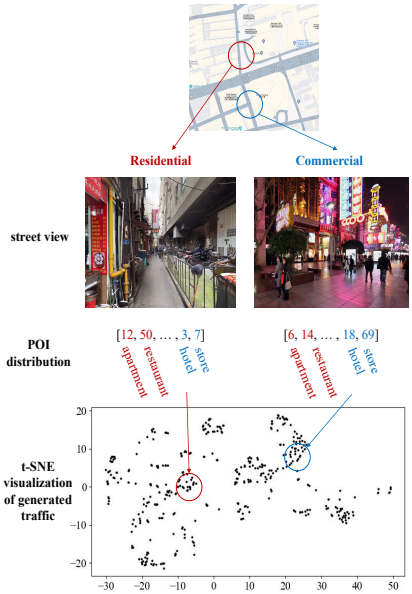


Fig. 3. Street views, POI distribution and t-SNE visualization of generated traffic of TG-SCR in two locations of Shanghai.

There are two important hyper-parameters that should be conducted, which are the combination coefficient α named in Eq (9) and the neighbor aggregator number k. We test the sensitivity of α by varying it from 0.001 to 0.1 using the regular experiment setting. Also, the sensitivity of k are tested by varying it from 1 to 10. The corresponding RMSE results are respectively shown in Figure 2 with blue lines. The blue shaded area in each line chart indicates the error range of the corresponding standard deviation. Among eight line chart in Figure 2, α is senstive merely in Futian dataset, and k is not sensitive in all the four dataset. We find that the performance of TG-SCR is generally stable with the change of α and k, which means the performance of TG-SCR is not sensitive to α or k.

5.5 Case Study

In case study, we select part of data in Shanghai dataset and visualize the generated traffic flow using t-SNE dimensionality reduction algorithm. The visualization results are shown in Figure 3. The two locations circled on the map with are closed in distance but different in POI distribution, representing residential area and commercial area respectively. It can be seen that despite close in graph, the generated traffic flows of nodes in these two locations have a significant difference, which shows that our model capture POI distribution relationship by employing supervised contrastive learning while end-to-end loss could not.

6 Conclusion

In this paper, we study on the traffic generation problem without using historical traffic data. We propose TG-SCR, a novel POI-based traffic generation model via supervised contrastive learning on reconstructed graph. First, TG-SCR uses a graph reconstruction module to reconstruct the original roadnet graph into a k-NN graph. After that, a POI supervised contrastive module is added to model the relationship between POI feature and traffic flow. To evaluate the performance of TG-SCR, we conduct experiments on four real-world datasets. Extensive experiments indicate that compared to other methods, TG-SCR can generate traffic flow with higher quality based on POI data in real-world datasets.

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